



Control Of Eye Movements



Objectives

After studying this chapter, students will be able to:

1. Explain the different types of eye movements and the 3 levels of eye movement control
2. Explain the role of CN III, IV and VI in the control of conjugated gaze
3. Name the deficits produced by lesion of the CN III, IV and VI and describe the symptoms and signs produced by these deficits. Recognize and differentiate them.
4. Name, localize and explain the function of the brainstem centers involved in the coordination of horizontal, vertical and vergence gaze.
5. Recognize and describe the deficits produced by lesions of the brainstem centers that control gaze
6. Indicate the location and function of the cortical centers concerned with the control of ocular movements
7. Indicate the commands centers that are important in the production of saccades and smooth pursuits
8. Recognize and describe the deficits produced by lesions of the cerebral areas that control ocular movements
9. Explain the pathways for light and accommodation reflexes

Eye Movements

OBJ. # 1

Conjugate Movements of the eyes

Move a peripheral image onto the FOVEA



Keep this image on the FOVEA even if the image is moving or the head is moving

Types Of Movements

- Horizontal Movements
 - Vertical Movements
 - Vergence Movements
- 

Eye Movement Components

- Saccades
- Smooth pursuits

OBJ. # 1

- Optokinetic reflex = Triggered by an object moving in the outside world
- Vestibulo-ocular reflex (VOR) = Stabilizes the eyes on a visual image during head and body movements.
- The VOR can be used to assess the integrity of the eye movement pathways in the brainstem in comatose patients
- NYSTAGMUS = a type of ocular movement with 2 characteristic phases:
 - a slow drift, followed by
 - a saccade in the opposite direction

Reflexive Eye Movements



OBJ. # 1

Nystagmus



OBJ. # 1

Optokinetic reflex





Control Of Eye Movements

OBJ. # 1

There are 3 levels of control

- The cranial nerve level of control
- The Gaze Centers level of control
- The cortex level of control

Cranial Nerve Level Control

**Extrinsic eye muscles
(skeletal muscles)**

- Oculomotor nerve, CN III
- Abducens nerve, CN VI
- Trochlear nerve, CN IV

**Intrinsic eye muscles
(smooth muscles)**

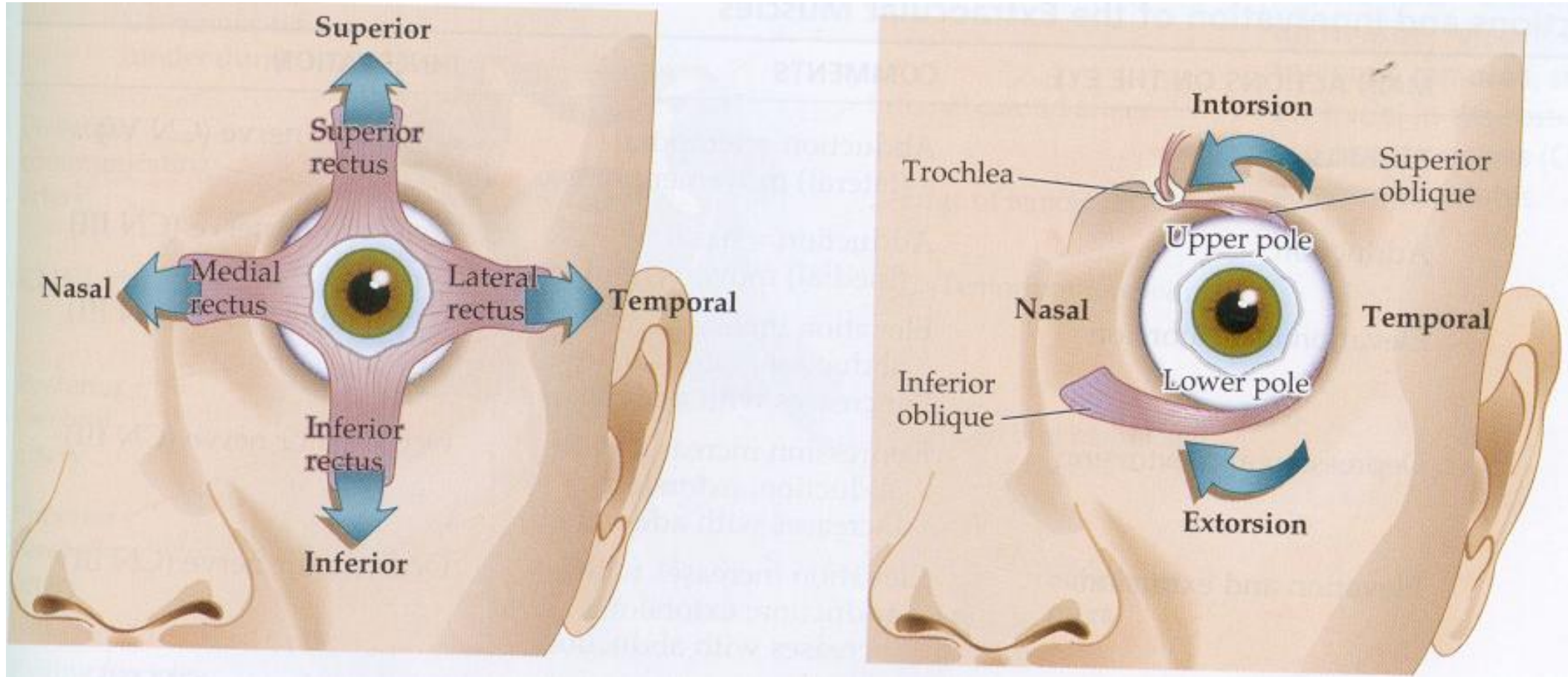


Autonomic innervation

- Pupillary constrictor and dilator
Ciliary muscles

Extrinsic Eye Muscles

OBJ. # 2

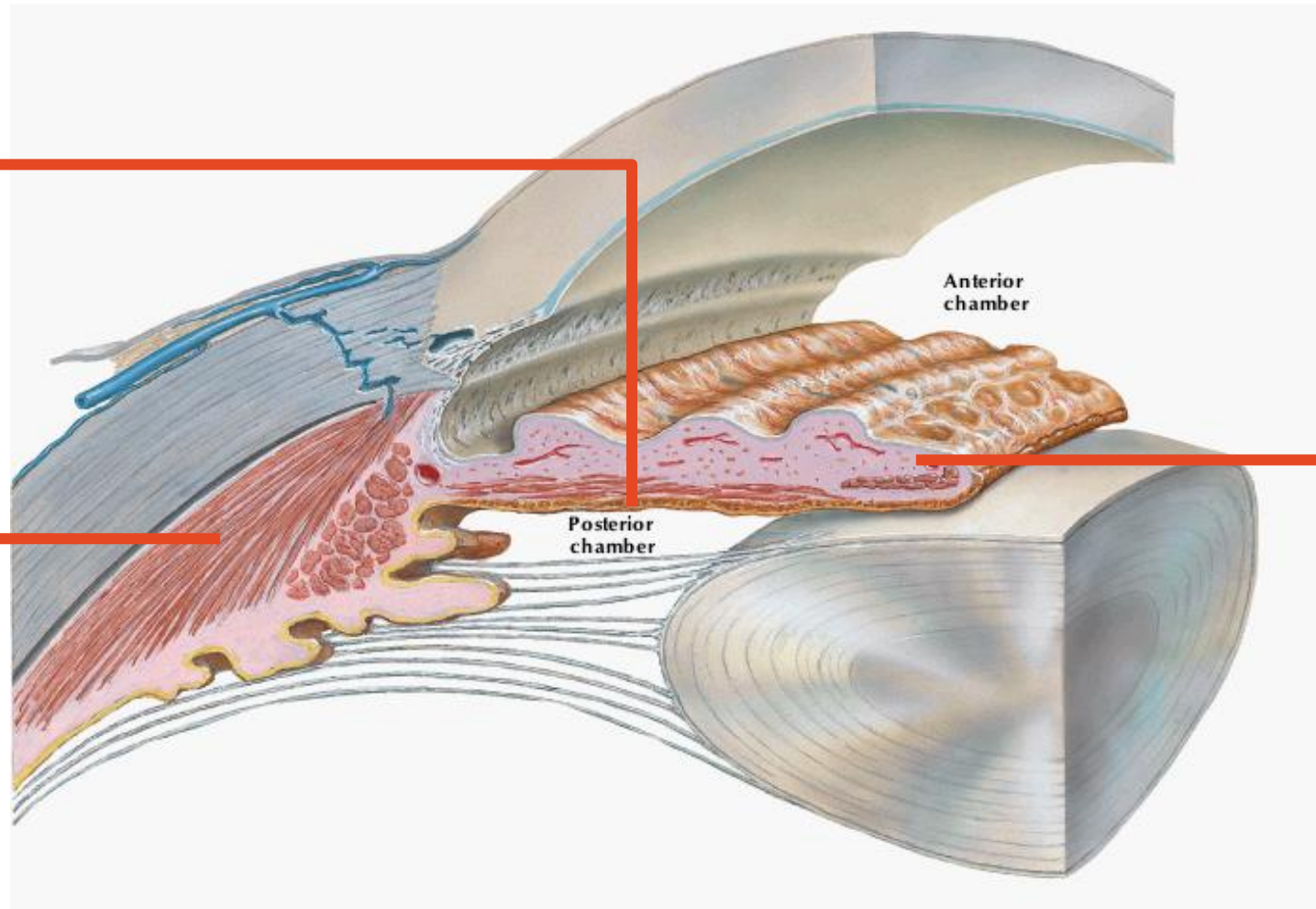


Intrinsic Eye Muscles

OBJ. # 2

**Dilator
Pupillary
Muscle**

**Ciliary
Muscles**



**Constrictor
Pupillary
Muscle**



Oculomotor, CN III

OBJ. # 2

Function

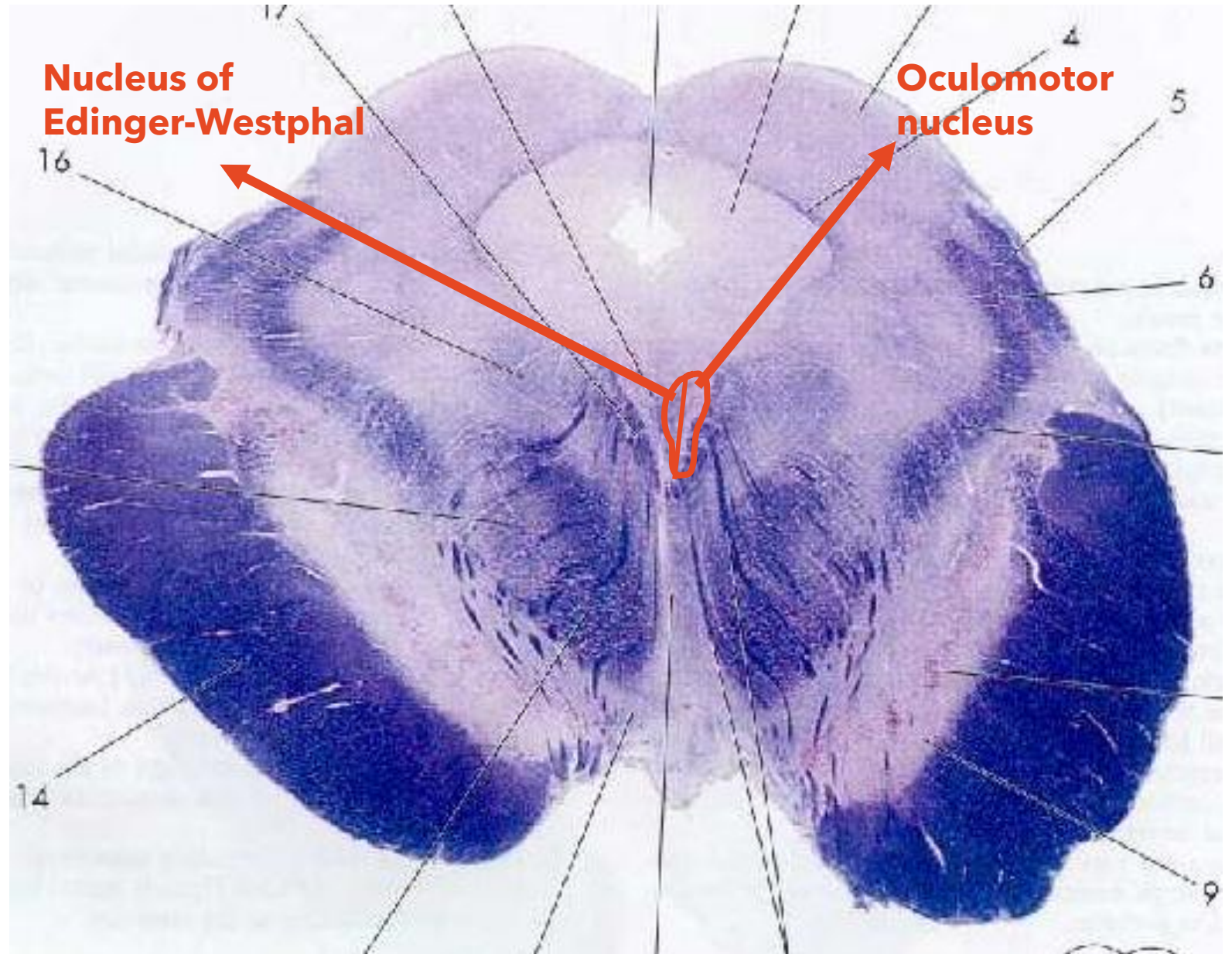
- Motor innervation of extrinsic muscles of the eye
- Parasympathetic motor innervation of smooth muscles of the eye

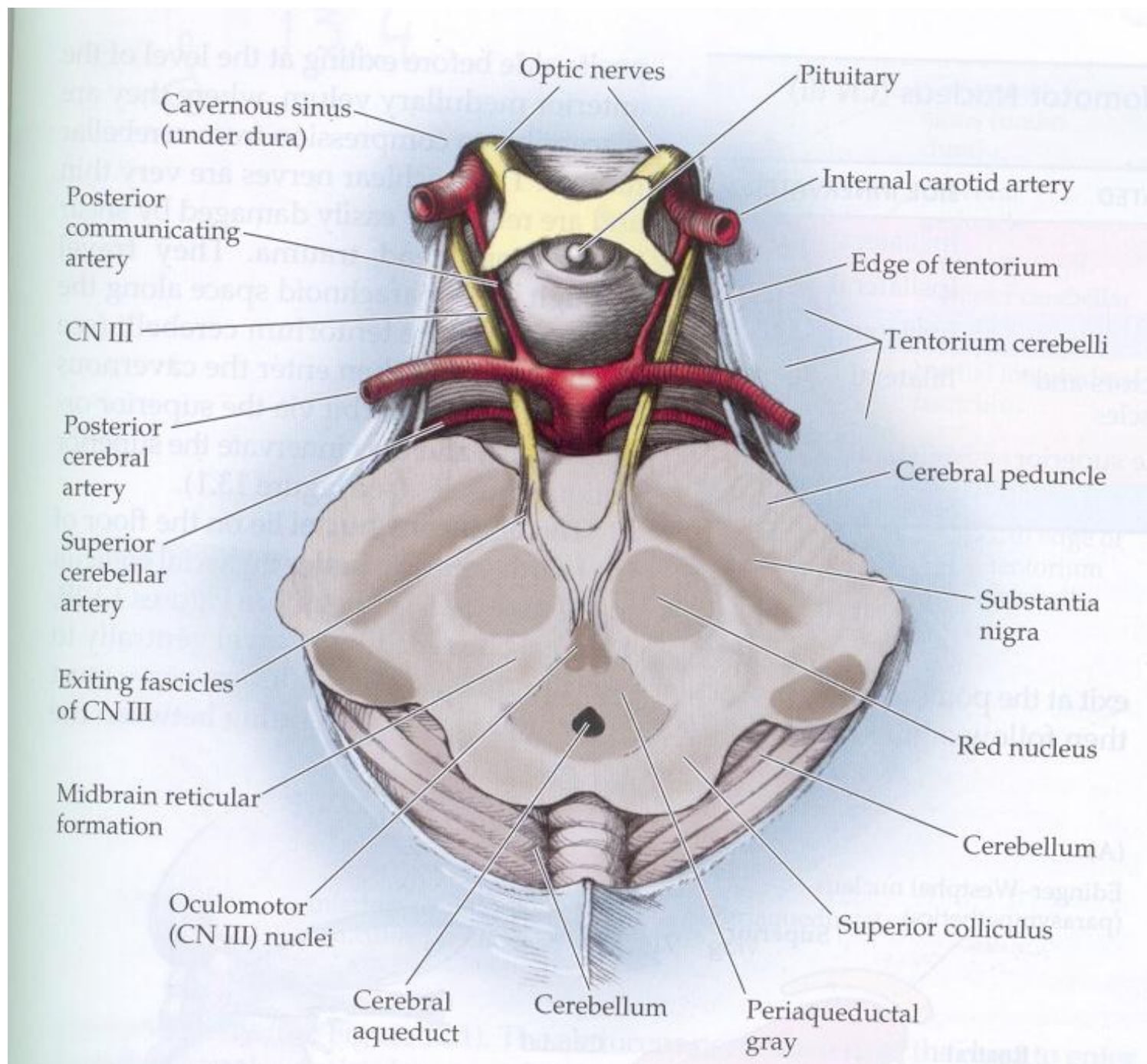
Nuclei

- Oculomotor Nucleus (rostral midbrain) → GSE
- Nucleus of Edinger - Westphal (parasympathetic, rostral midbrain) → GVE

Oculomotor Nerve

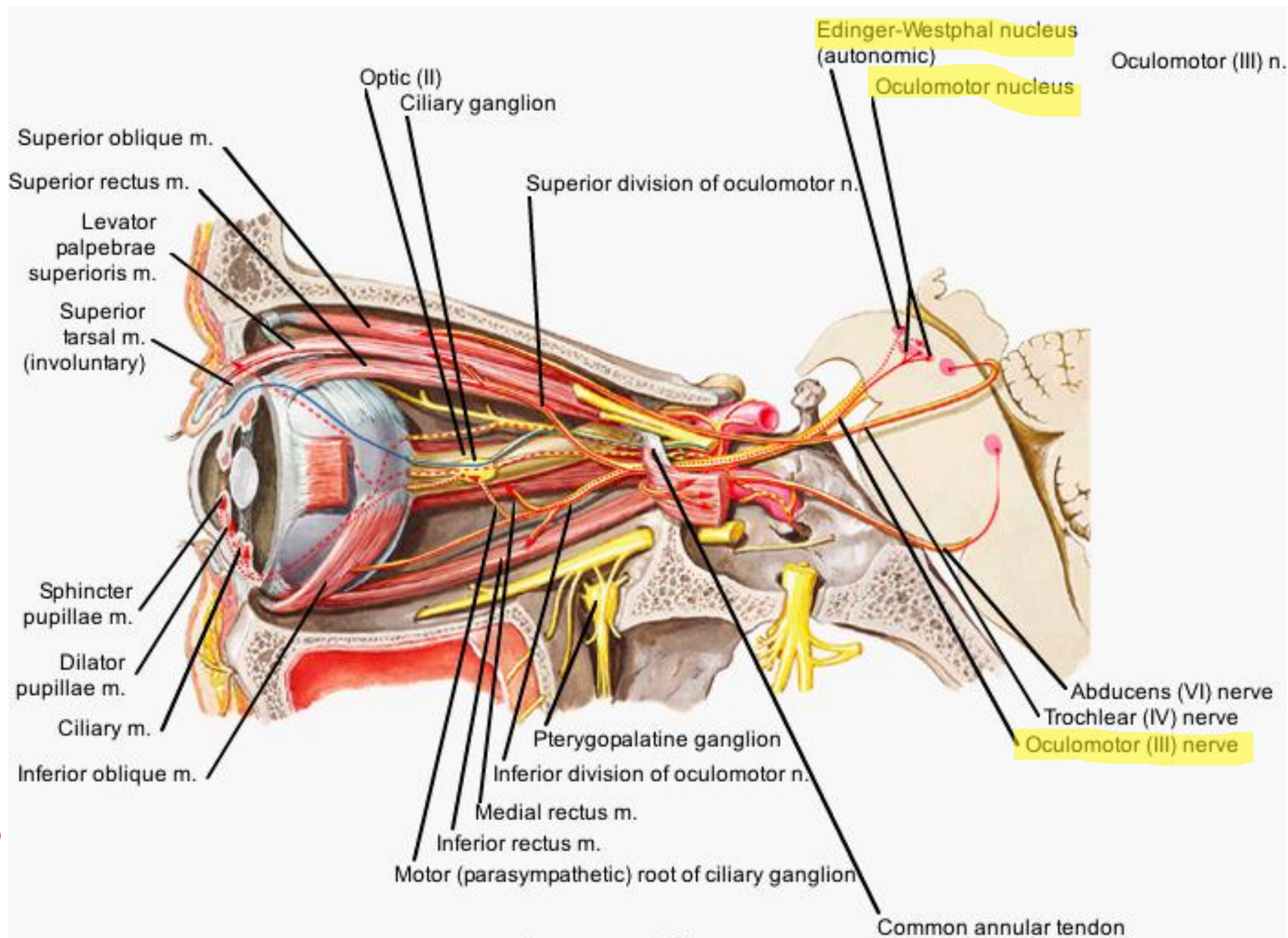
OBJ. # 2





OBJ. # 2

OBJ. # 2



Oculomotor Nerve, CN III

Reflexes:

- Light and accommodation reflexes
 - Afferent limb: optic nerve
 - Efferent limb: oculomotor nerve

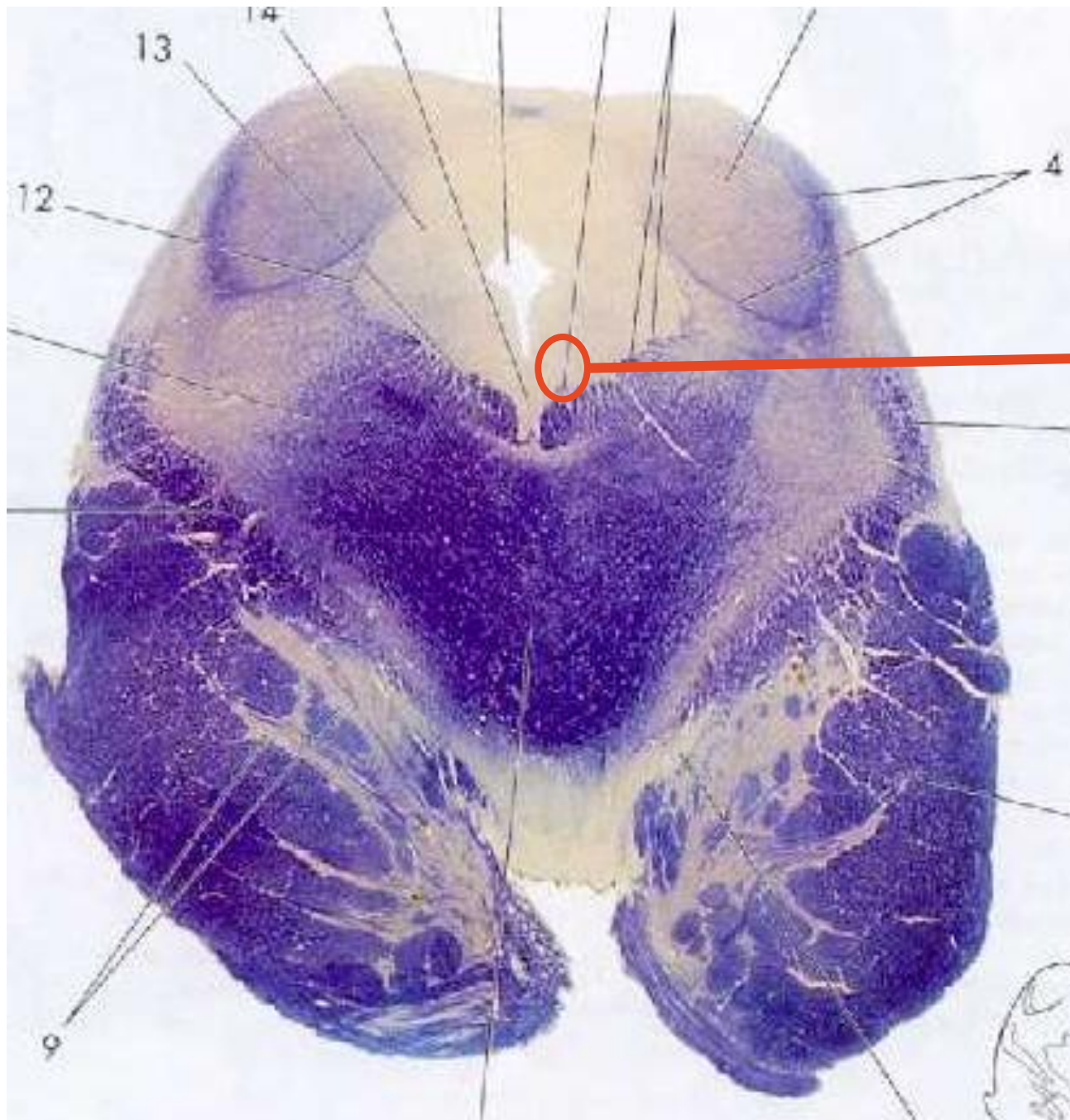


Trochlear Nerve, CN IV

Function → Motor innervation of superior oblique muscle of the eye

Nucleus → Trochlear nucleus (caudal midbrain) → GSE

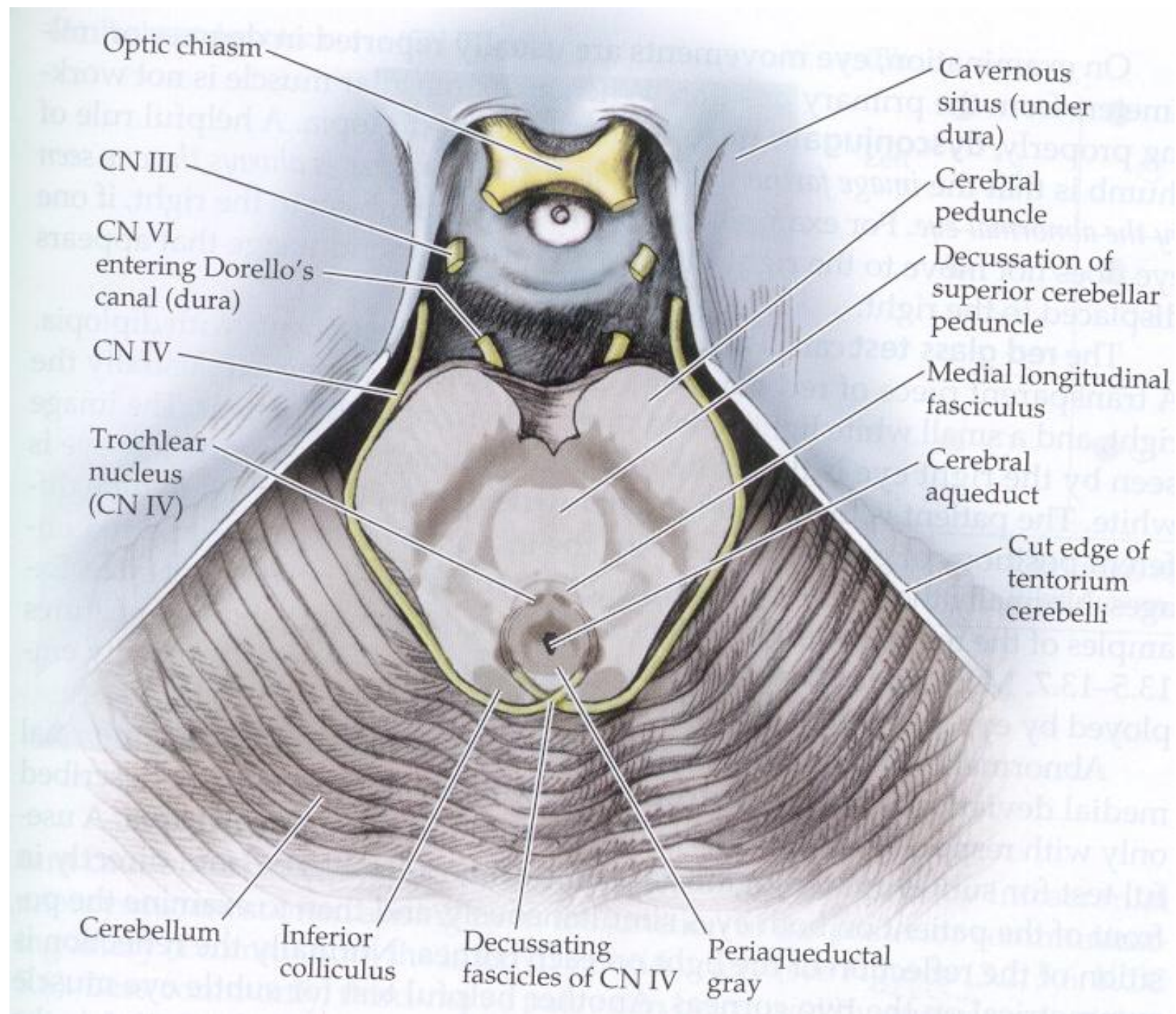
OBJ. # 2



**Trochlear
nucleus**

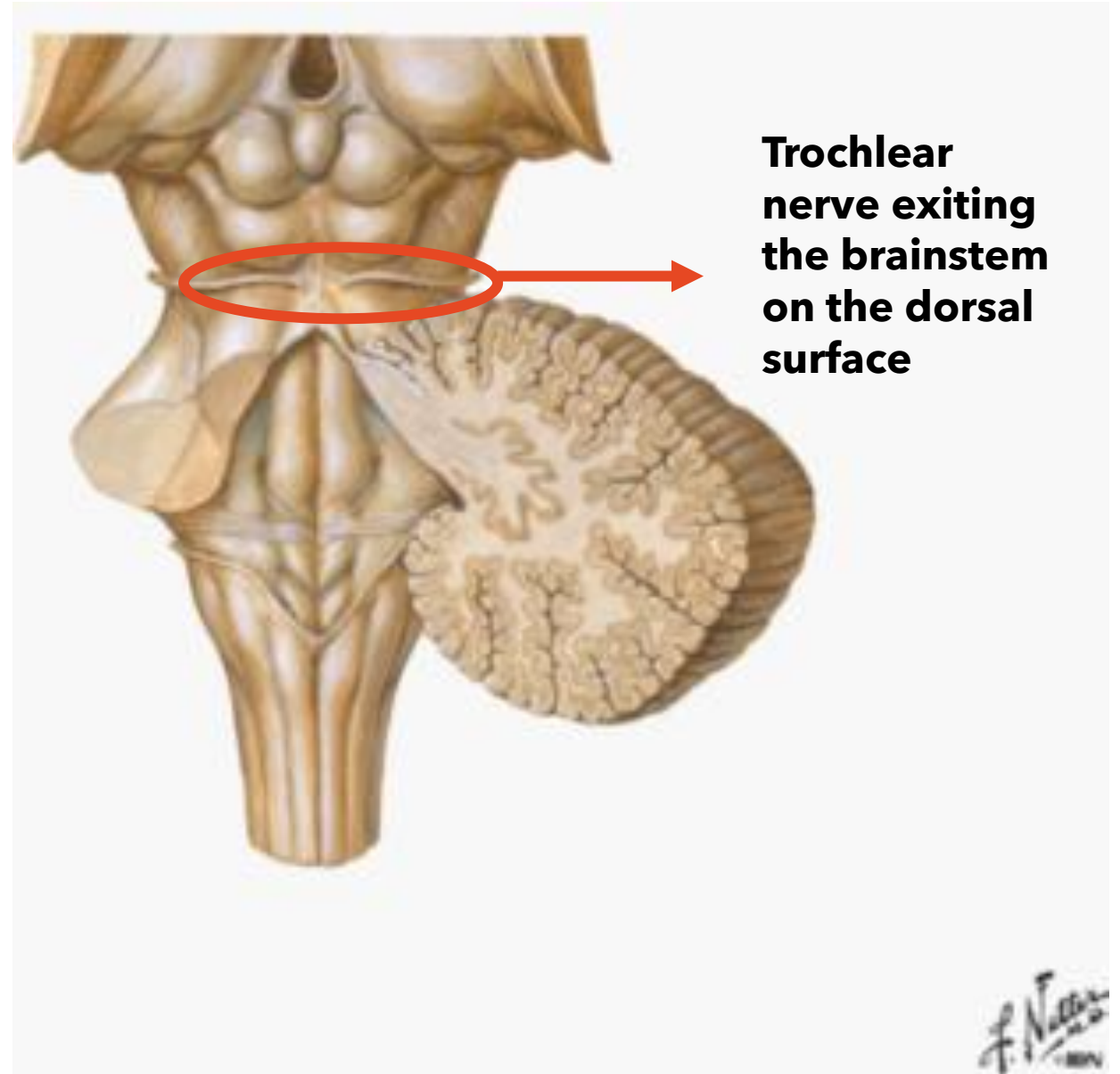


OBJ. # 2



Anatomy of the Fourth Ventricle – Posterior view

OBJ. # 2

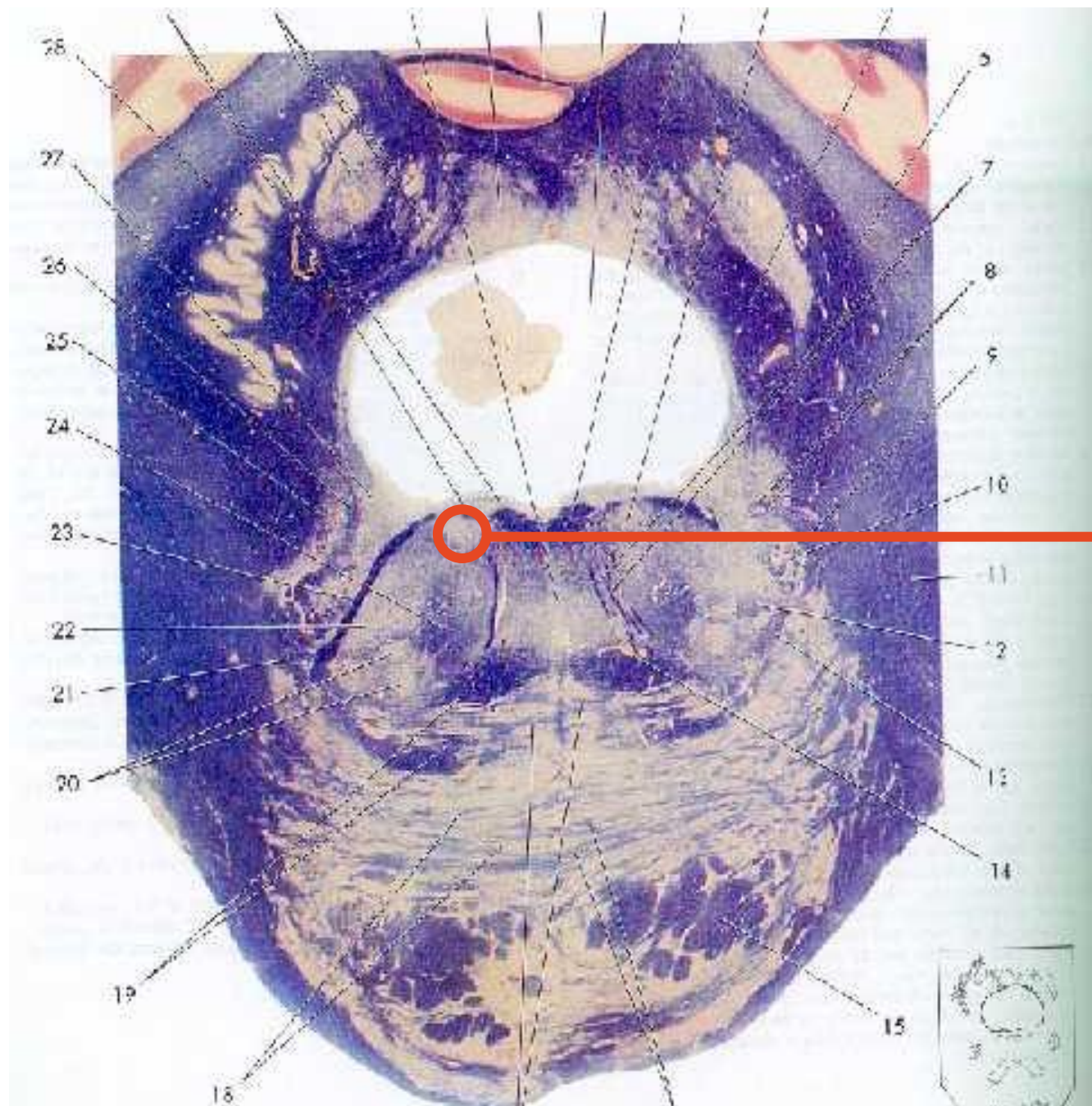


Abducens Nerve, CN VI

Function → Motor innervation of the lateral rectus muscle of the eye

Nucleus → Abducens nucleus (caudal pons) → GSE

OBJ. # 2

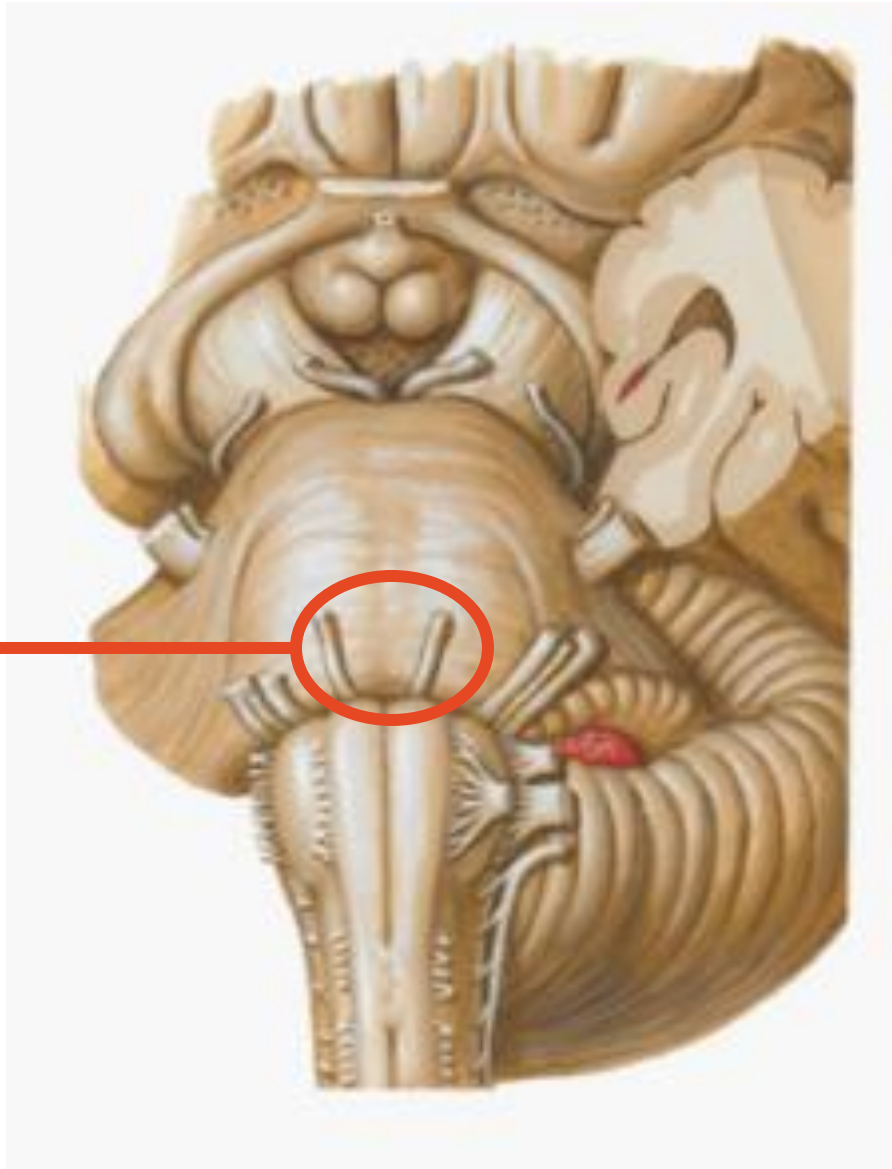


Abducens nucleus



OBJ. # 2

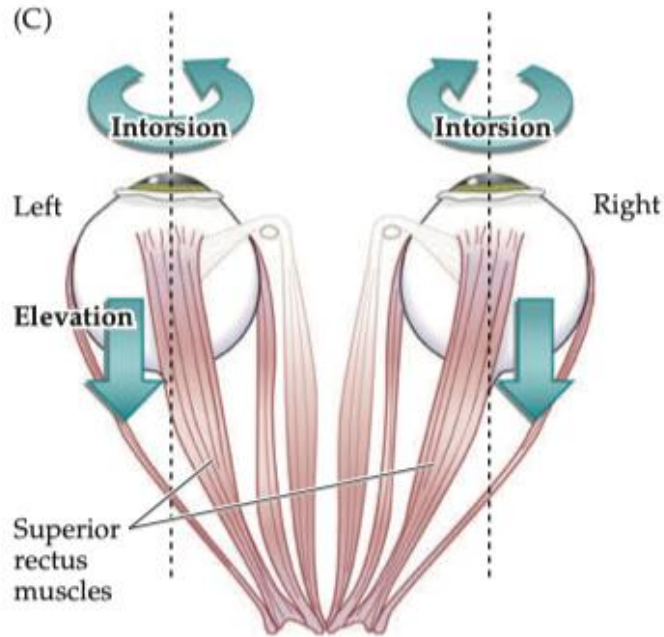
Abducens Nerve



**Brain Stem
Surface
Anatomy:
Anterior View**

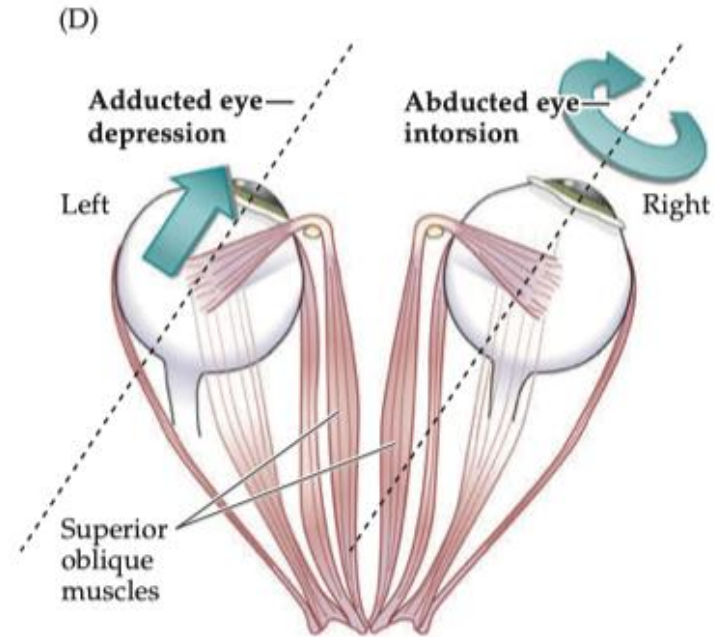
Oculomotor Nerve

OBJ. # 3



MUSCLE

Medial rectus
Superior rectus
Inferior rectus
Inferior oblique



MAIN ACTION

Adduction
Elevation and intorsion
Depression and extorsion
Elevation and extorsion

INNERVATION

Oculomotor nerve, CN III
Oculomotor nerve, CN III
Oculomotor nerve, CN III
Oculomotor nerve, CN III

Lesion of CN III or its nucleus produces paralysis of all muscles innervated by it. The only movements left are:

- Some abduction
- Some depression and intorsion



- The eye is in a **Down And Out** position at rest
- DIPLOPIA = Double vision
- Ptosis = Paralysis of the levator palpebrae superioris

If there is involvement of parasympathetic fibers



- Anisocoria = Pupillary asymmetry and
- Mydriasis = unresponsive dilated pupil

Oculomotor Palsy

Oculomotor Nerve Deficits



The R eye is in a down and out position

Oculomotor Palsy



Symptoms consistent with palsy of third cranial nerve

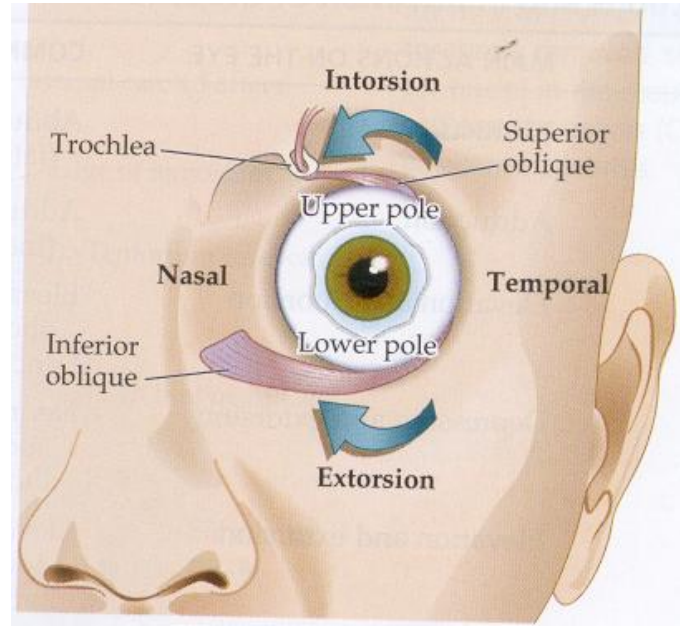
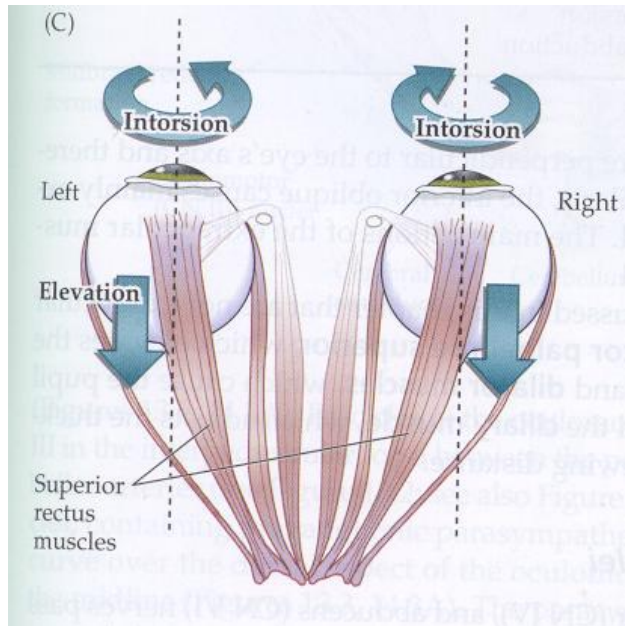
OBJ. # 3

A) Complete right upper lid ptosis in primary position and B) complete limitation of adduction of the right eye on left gaze.



Trochlear Nerve

OBJ. # 3



MUSCLE

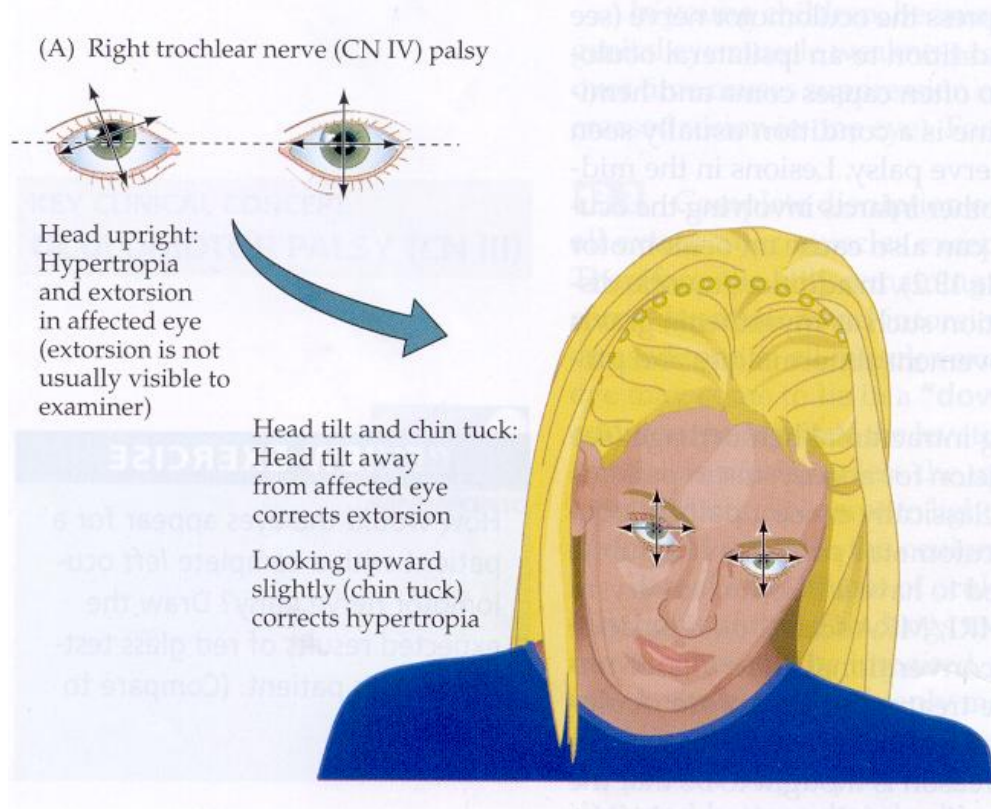
Superior oblique

MAIN ACTION

Depression and
intorsion

INNERVATION

Trochlear nerve, CN IV



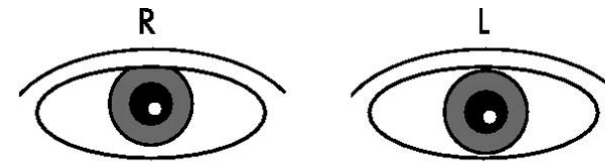
Trochlear Palsy

Trochlear Nerve Deficits

- Extorsion of the eye
- Hypertropia = The eye is in a higher position at rest
- Vertical diplopia that worsens when the patient looks down and toward the nose
- Patients tend to look up and tilt the head away from the affected eye

OBJ. # 3

Trochlear Palsy



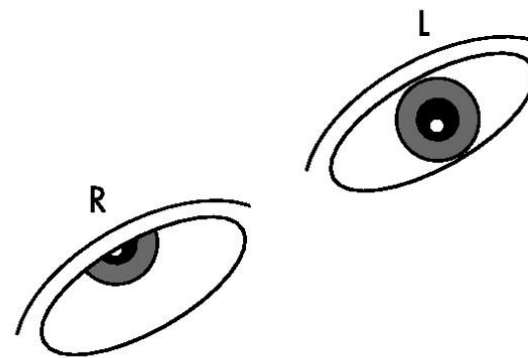
Primary position



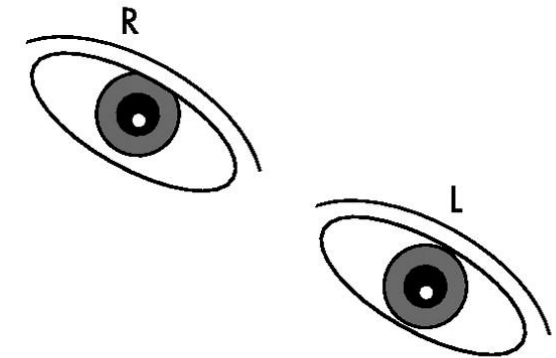
Right gaze



Left gaze



Head tilt right



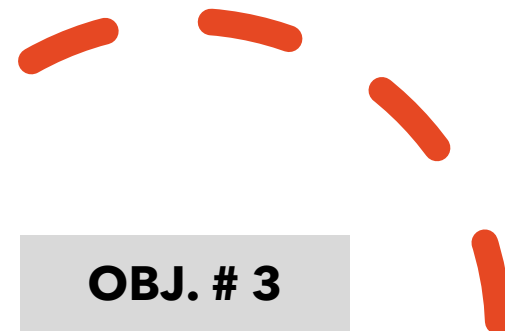
Head tilt left

OBJ. # 3



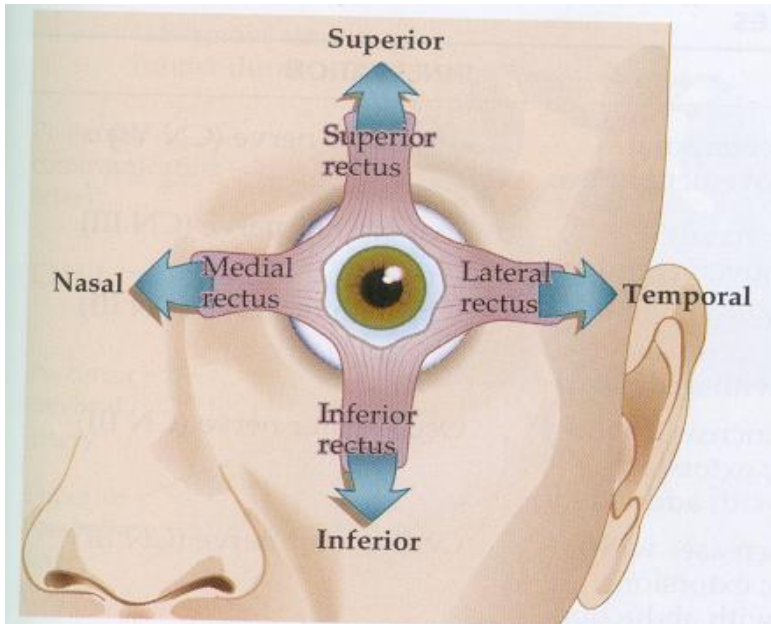
Trochlear Palsy





Abducens Nerve

OBJ. # 3



MUSCLE

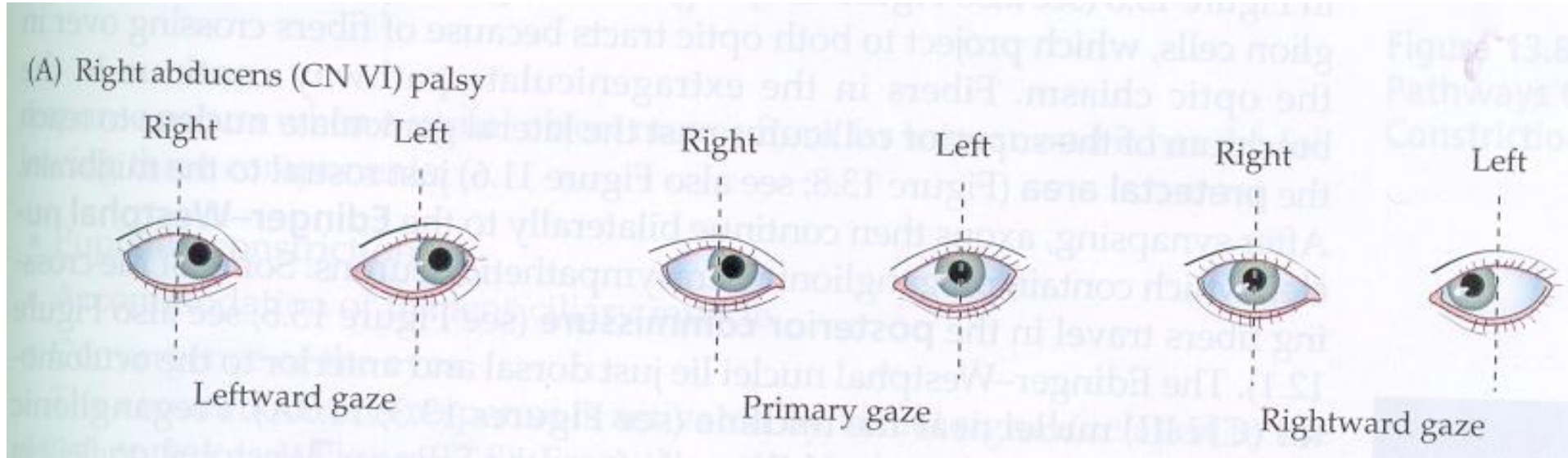
Lateral rectus

MAIN ACTION

Abduction

INNERVATION

Abducens nerve, CN VI



A lesion of the abducens nerve fibers will produce:

- Horizontal diplopia
- Esotropia of the affected eye = abnormal medial deviation of the eye
- Diplopia worsens with abduction of affected eye

Some Patients tend to turn the head toward the affected eye

OBJ. # 3

Abducens Palsy

Abducens Nerve Deficits



Source: Comp Ophthalmol Update © 2006 Comprehensive Ophthalmology Update, LLC

OBJ. # 3

Affected eye is medially deviated
in primary position

Abducens Palsy

Abducens Palsy



Control Of Gaze

Gaze Centers level of control

The Brainstem





Supranuclear Control Of Eye Movements

Brainstem control

OBJ. # 4

Horizontal gaze center

Para-median Pontine Reticular Formation (PPRF) + Abducens nucleus

Vertical and vergence
gaze centers



Midbrain reticular
formation

Other control centers

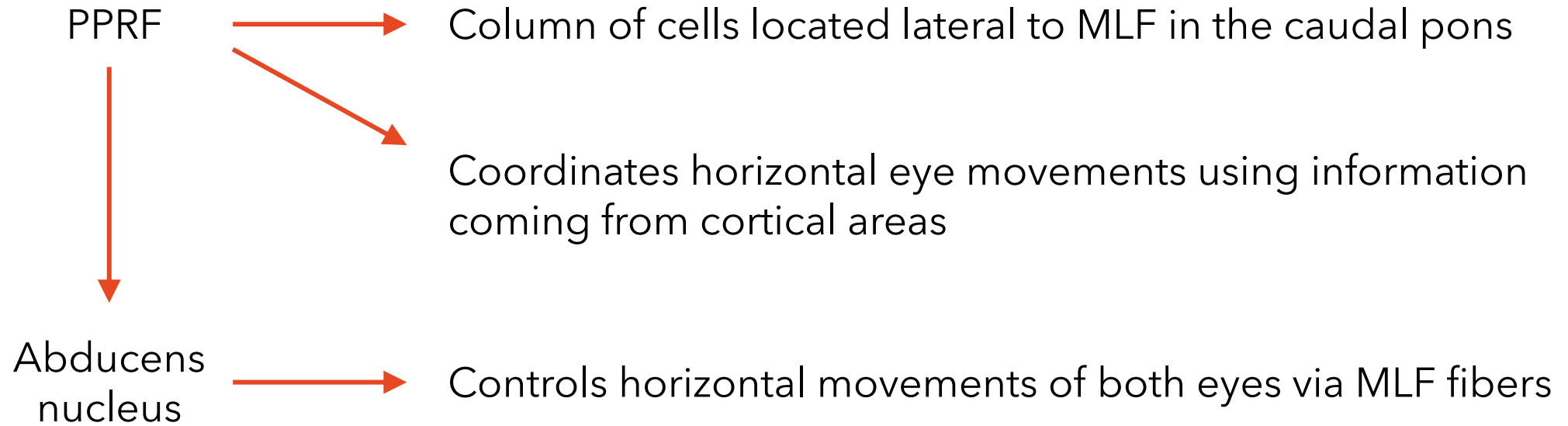


- Superior colliculus
- Cerebellum
- Vestibular nuclei

The Gaze Centers

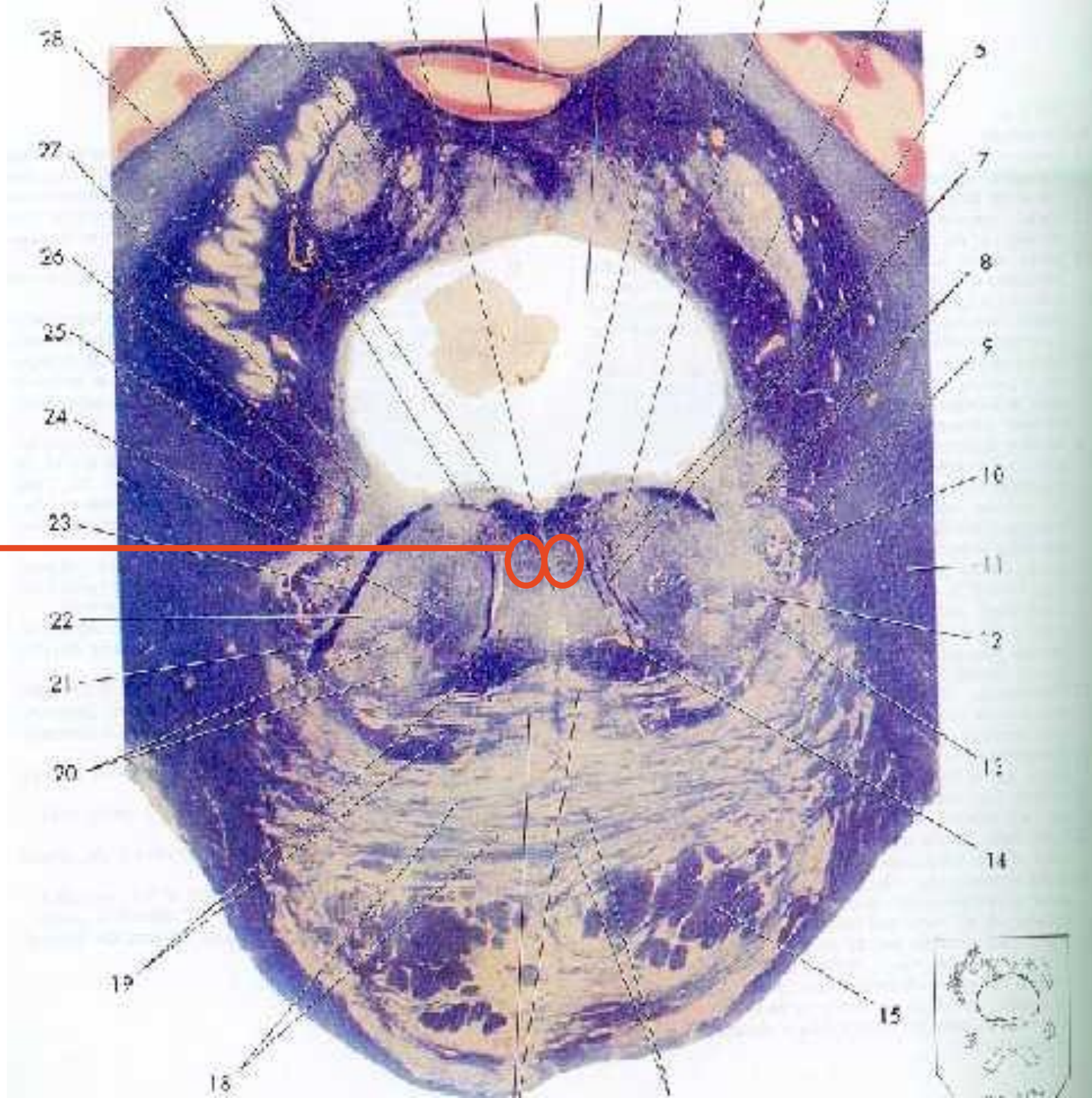
OBJ. # 4

Horizontal gaze centers



OBJ. # 4

PPRF

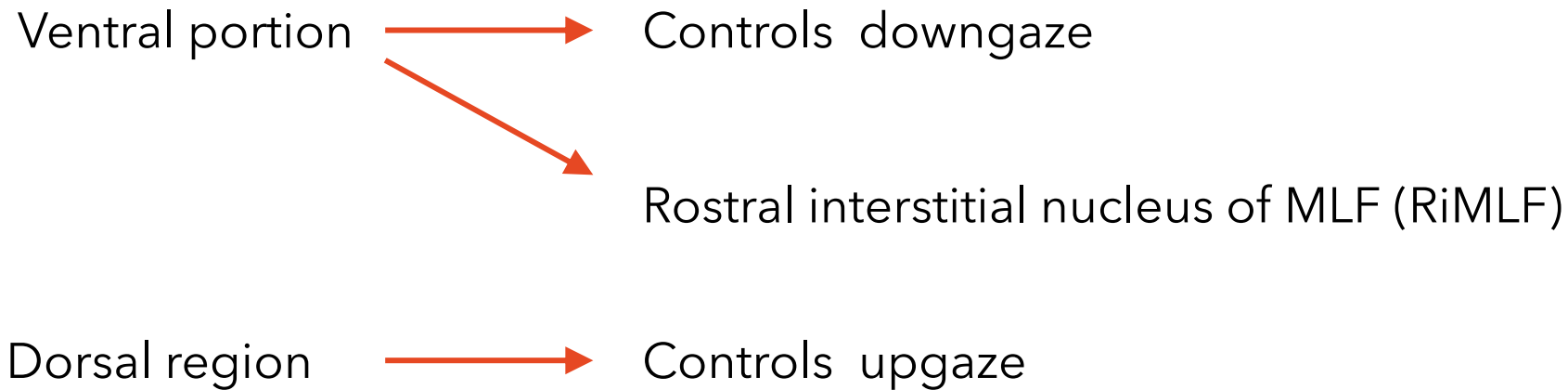


The Gaze Centers

OBJ. # 4

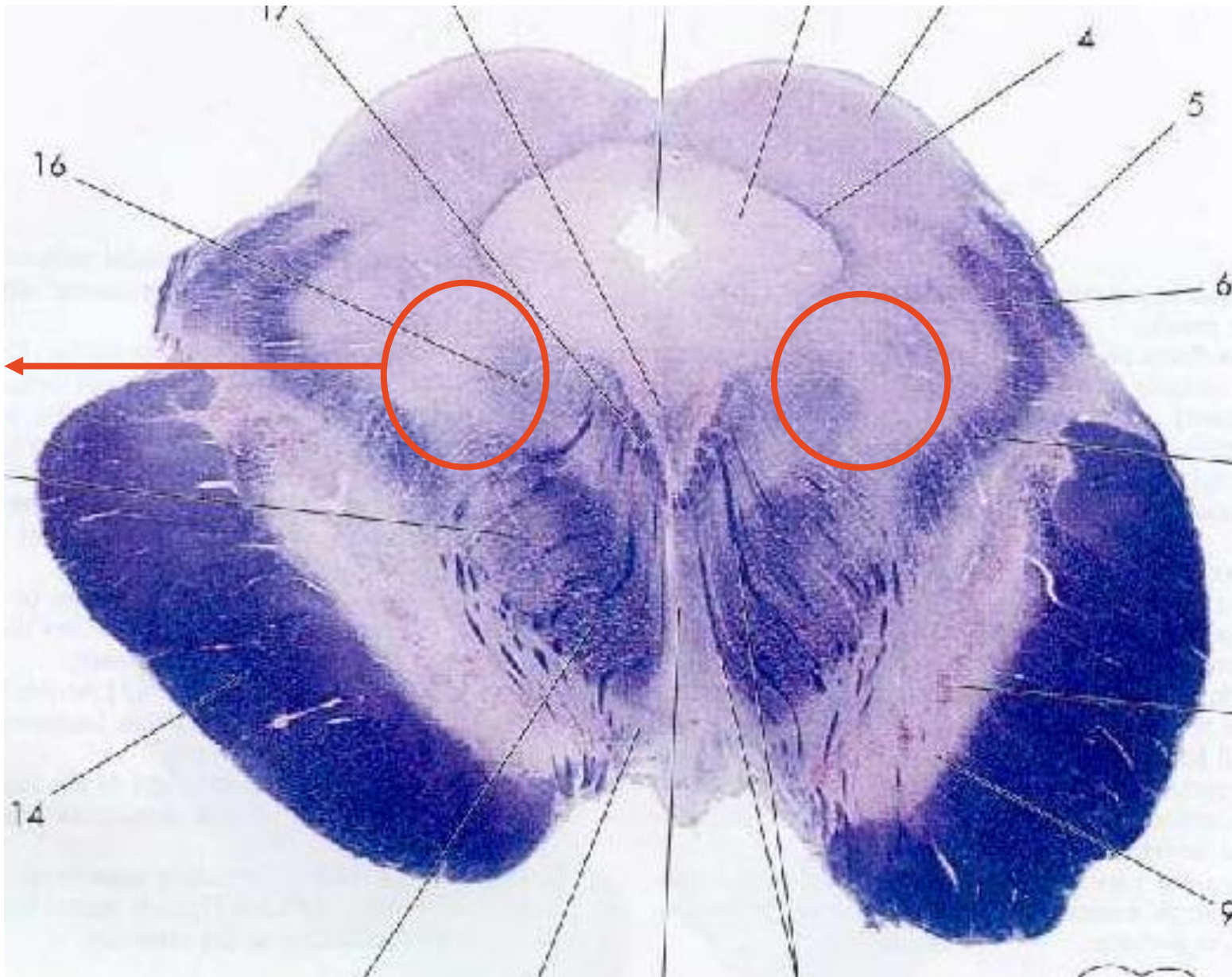
The rostral midbrain reticular formation controls vertical and vergence eye movements

Vertical gaze center

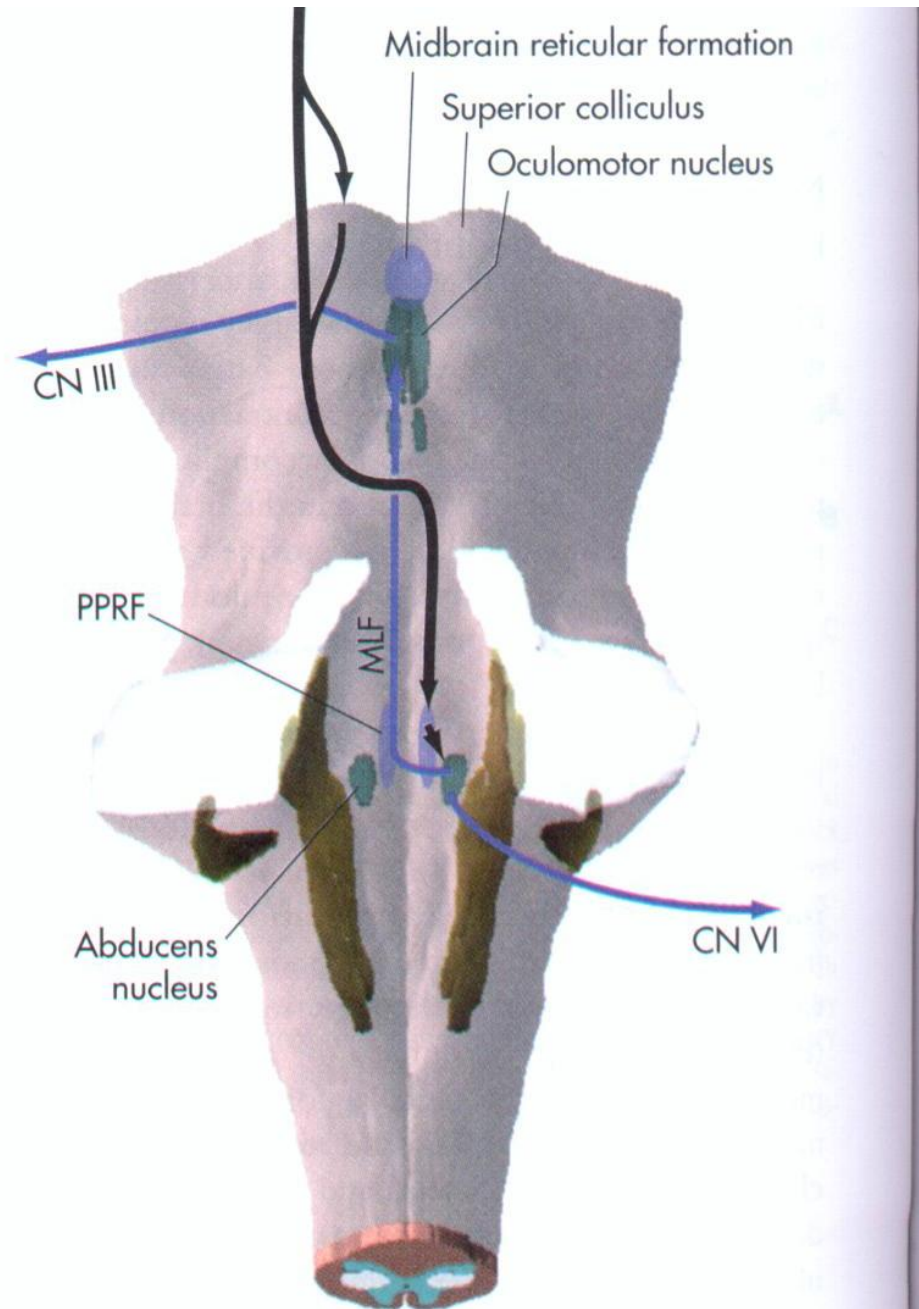


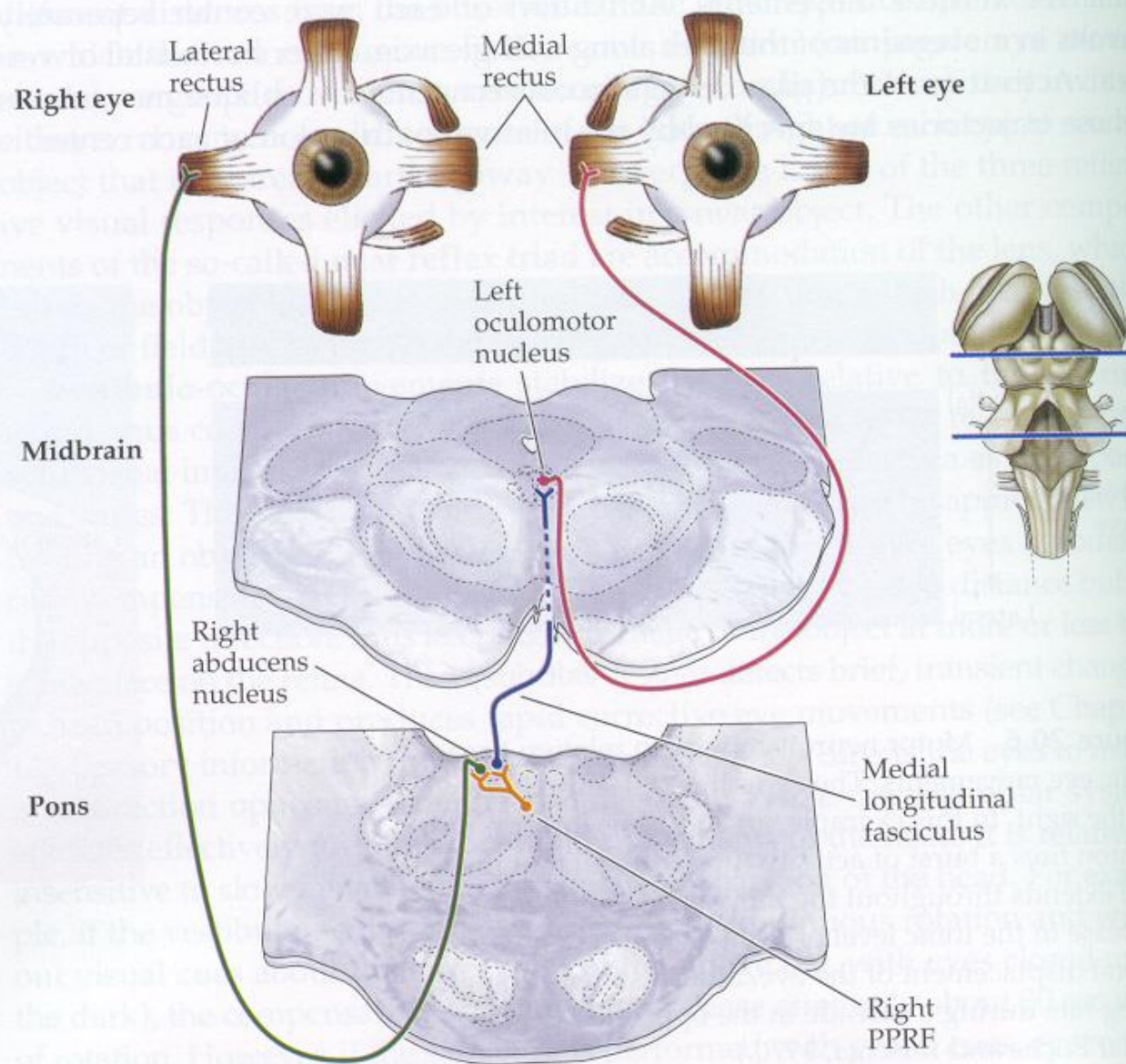
OBJ. # 4

**Vertical and
Vergence gaze
centers**



OBJ. # 4



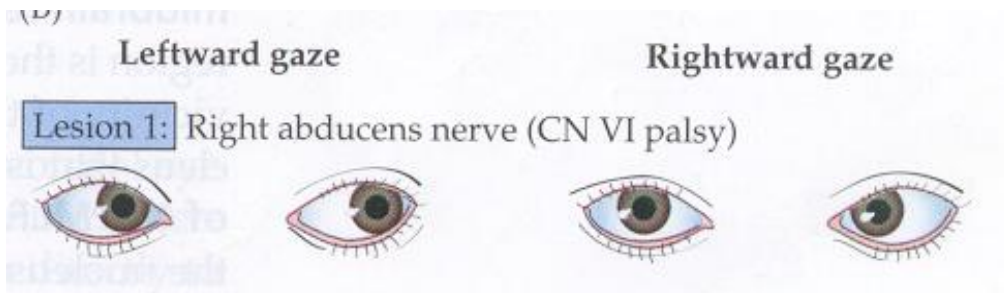


**Diagram showing
the control of
horizontal gaze**

Brainstem Lesions Affecting Horizontal Gaze

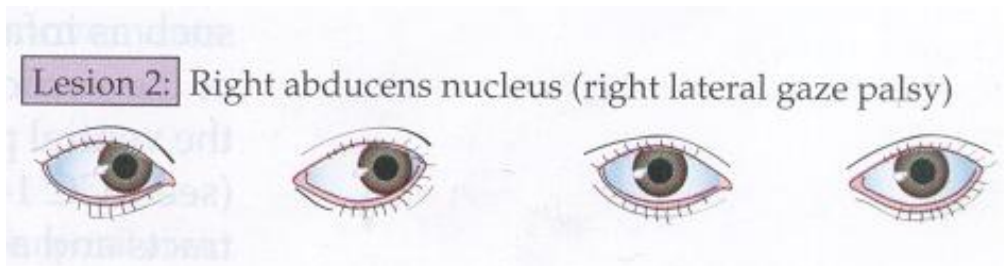
OBJ. # 5

Lesions of the abducens nerve: ABDUCENS PALSY

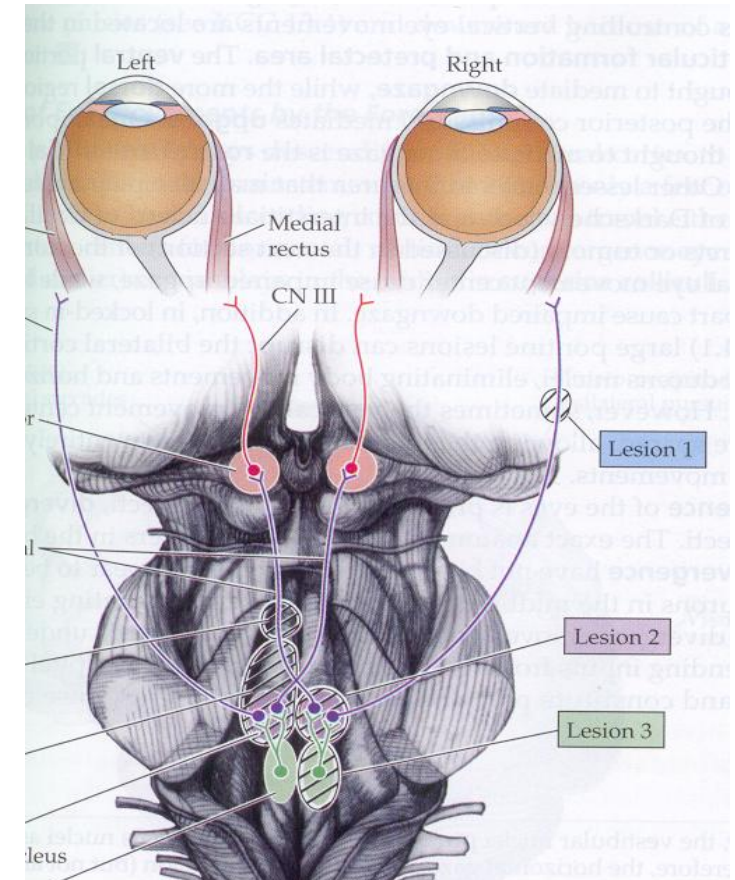


**Abducens Palsy =
one eye affected**

Lesions of the abducens nucleus or PPRF- lateral gaze palsy



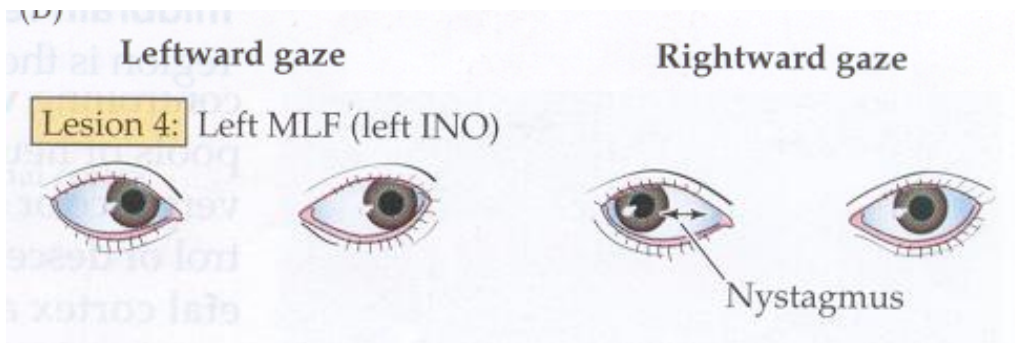
**Lateral gaze palsy =
both eyes affected**



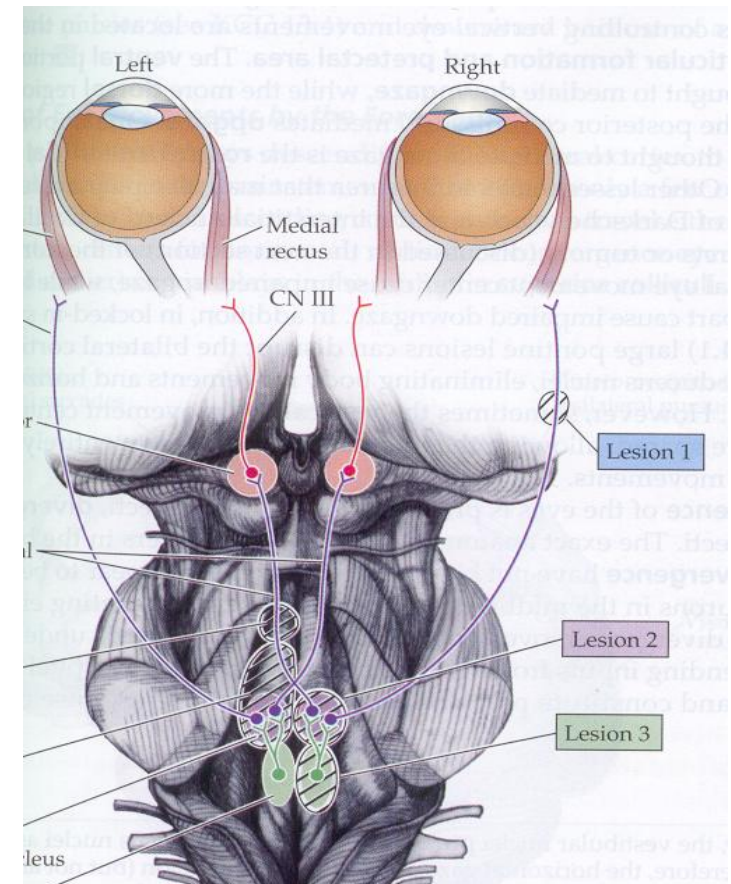
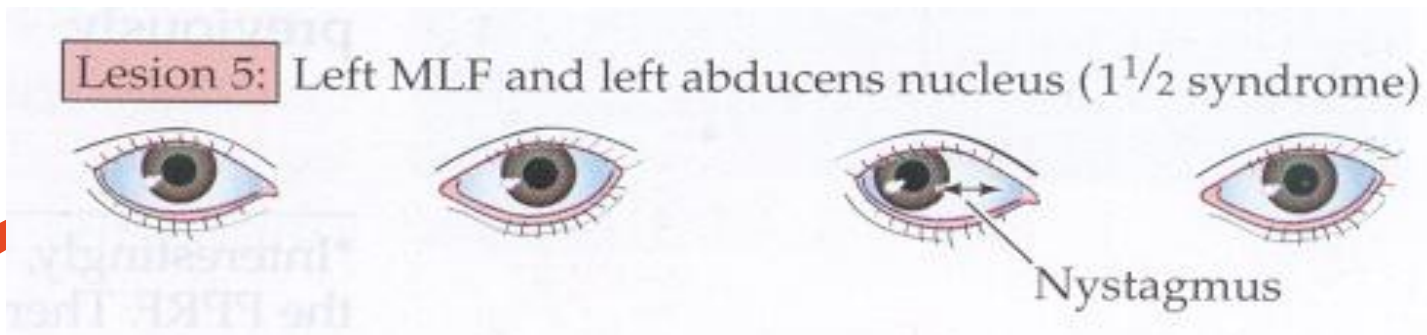
Brainstem Lesions Affecting Horizontal Gaze

OBJ. # 5

Lesions of the MLF - Internuclear Ophthalmoplegia



Lesions of the MLF + abducens nucleus ipsilaterally:
one-and-a-half syndrome



Internuclear ophthalmoplegia video

OBJ. # 5



Bilateral Internuclear ophthalmoplegia

OBJ. # 5



**Eight and a half syndrome:
One and a half syndrome plus facial nerve palsy**

OBJ. # 5



Supplement to *Arch Neurol*. doi:10.1001/archneurol.2011.2185. © AMA.

Brainstem Lesions Affecting Vertical Gaze

OBJ. # 5

Parinaud's Syndrome – Produced by a compression of the dorsal midbrain affecting the vertical and vergence gaze centers

Main symptoms are:

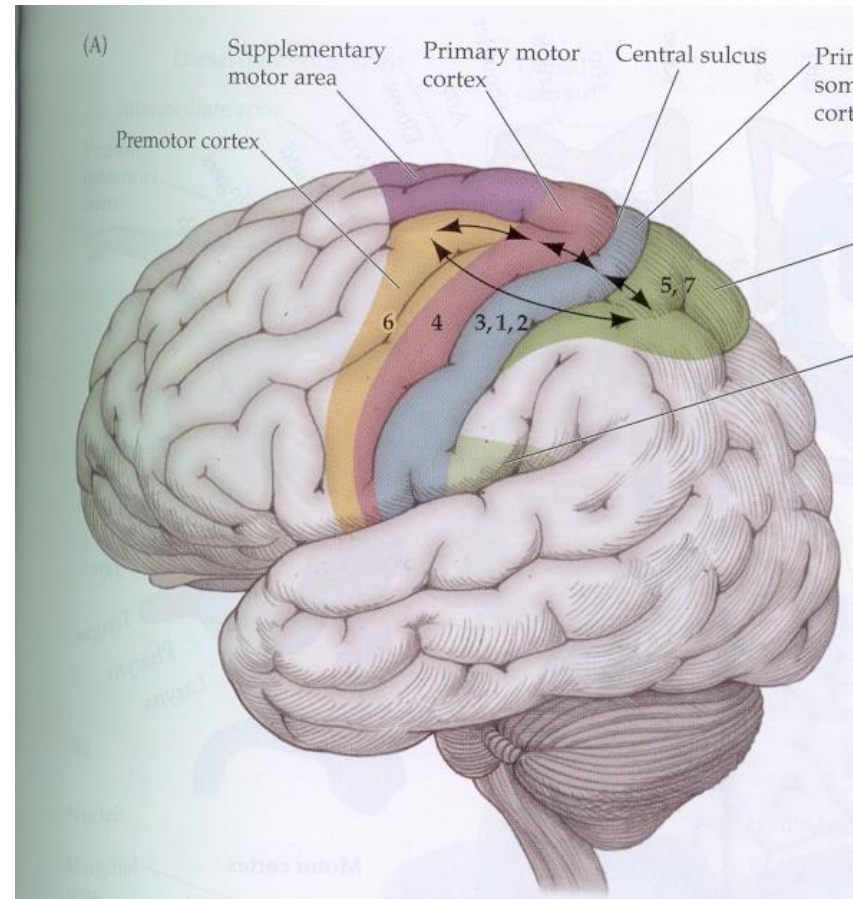
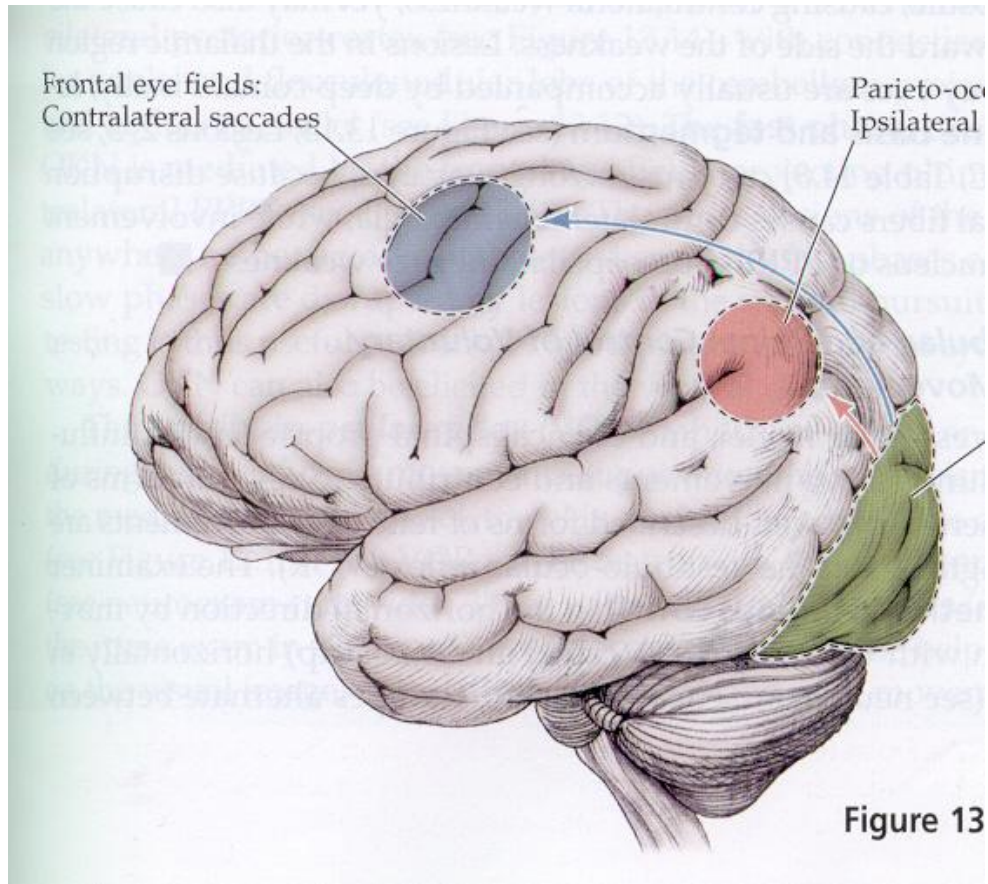
- Impair vertical gaze
- Pupillary light-near dissociation
 - Pupils unresponsive to light, but constrict with convergence
- Convergence-retraction nystagmus
- Collier's sign – lid retraction



Supranuclear Control Of Eye Movements

OBJ. # 6

- Cortical areas:
 - Frontal eye fields
 - Parieto-occipito-temporal cortex



OBJ. # 6

Frontal And Parietal Eye Fields



Control Of Saccade Movements

OBJ. # 7

Frontal Eye Fields

Initiate saccades in the contralateral direction via projections to the ipsilateral superior colliculus and the contralateral PPRF

Parieto-occipito-temporal
cortex provides visual
Input to the Frontal Eye
Fields

Control Of Smooth Pursuit Movements

OBJ. #7

Parieto-occipito-temporal cortex



Control of ipsilateral smooth pursuit movements



Cerebellum

Vestibular Nuclei



PPRF / vertical gaze center

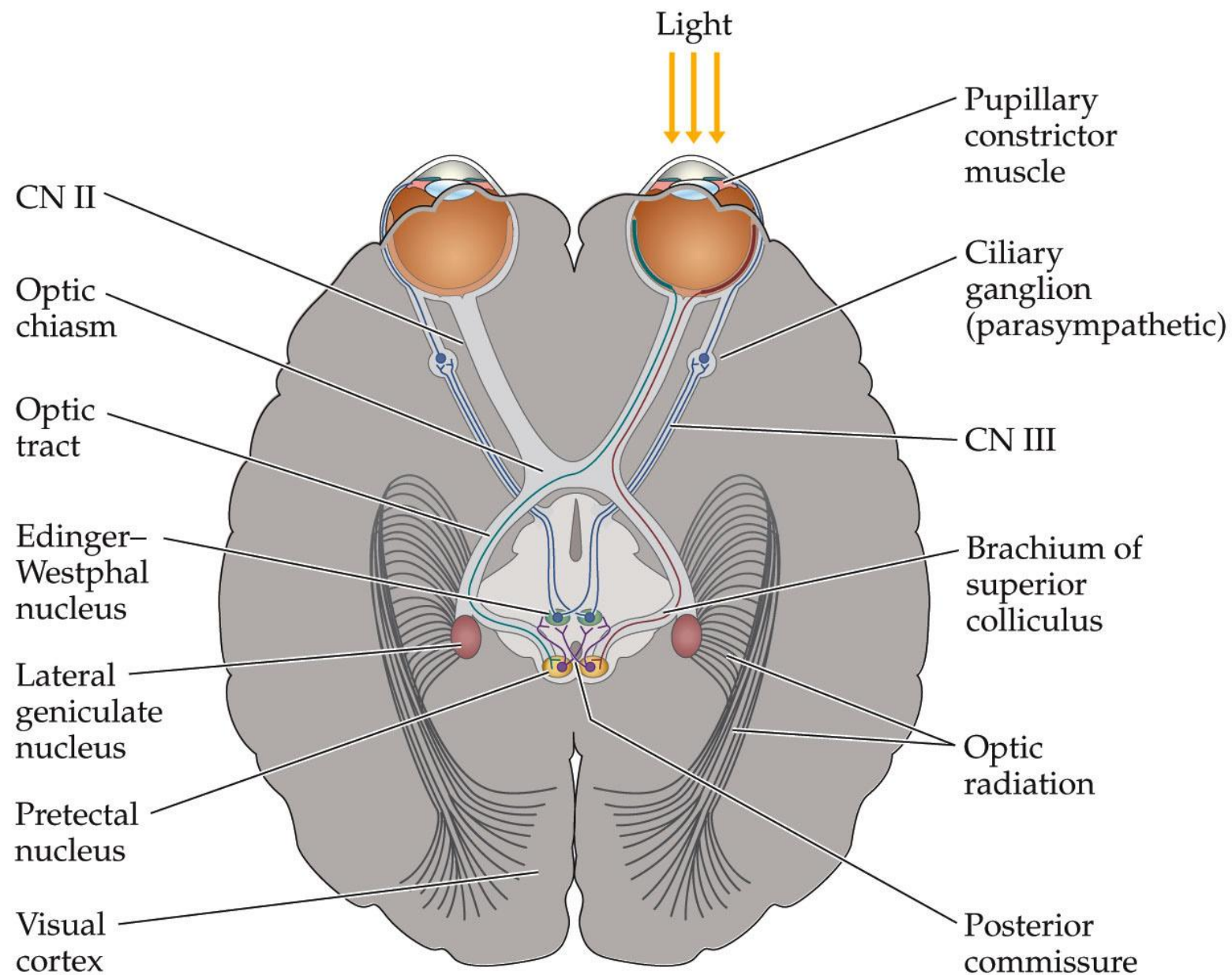


Eye Movement deficits due to cerebral cortex damage

OBJ. # 8

- Damage to the frontal eye fields in the motor cortex produces:
 - Impaired saccades to the contralateral side
 - “Gaze preference” towards the side of lesion
 - Usually there is weakness or hemiplegia on the contralateral side of the body and face
- Damage to the parietal eye fields produces:
 - Impaired smooth pursuits to the ipsilateral side

OBJ. # 9



**Light
Reflex**

Accommodation

OBJ. # 9

There are 3 components in accommodation:

- Convergence of the eyes
- Constriction of the pupils
- Accommodation of the lens

